

# DataLink & Product Streamer

## Software Verification Report

Technical briefing | Quan Guo | Updated 22 September 2026

### 1. Software and Versions

**Objective.** Verify the software path from a synthetic data identifier (DID), through DataLink discovery, to Product Streamer delivery and independent file-integrity checks. The laboratory uses official SKA service code with simulated supporting services.

| Component              | Software, version and role                                                                                                                                                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DataLink               | Official <a href="#">ska-src-dm-datalink</a> , <b>0.1.5</b> . Resolves DIDs and returns VOTable links and service descriptors. Commit: <code>997dfb03944b9c44283411264ca1c07b48b03a8e</code> .                                  |
| Product Streamer       | Official <a href="#">ska-src-dm-product-streamer-api</a> , <b>0.1.2</b> . Delivers single files and TAR datasets. Commit: <code>27df8cb247105faf67a6d278dd9819528f211dd3</code> .                                               |
| Server environment     | Recorded baseline: <b>Rocky Linux 10.2</b> , <b>Python 3.13.15</b> , three separate Python environments, systemd process supervision.                                                                                           |
| Service dependencies   | Official services: <b>FastAPI 0.124.4</b> , <b>Uvicorn 0.34.3</b> . Portal: <b>FastAPI 0.141.1</b> , <b>Uvicorn 0.53.0</b> . HTTP client: <b>HTTPX 0.28.1</b> . Full dependency snapshots are included in the repository.       |
| Laboratory integration | Custom Python portal, password gateway, process supervisor and deterministic fixture generator, with HTML/CSS/JavaScript UI. This code has no separate semantic release version; Git history identifies the published snapshot. |
| Python tests           | Custom external client <code>test_public_e2e.py</code> : <b>Python 3.9+</b> , standard library only. Additional scripts cover the internal suite, gateway and client regressions.                                               |

**Test environment.** <http://64.176.188.89:18080>. DataLink and Streamer run on server loopback ports 18081 and 18082 behind the laboratory gateway. The official application source is unmodified; configuration directs dependencies to local fixtures.

**Simulation boundary.** DMAPI, IAM, PAPI and colocated-service discovery are simulated. No real SCAPI, Rucio, StoRM or federated SRCNet service is connected. Four deterministic files are used: FITS, text, JSON and a 1 MiB binary. The dataset contains the first three files. Public HTTP is unencrypted; use synthetic data and a disposable laboratory password only.

**Evaluation basis.** [IVOA DataLink 1.1](#), [VOTable 1.4](#), [HTTP Semantics / RFC 9110](#) and [SHA-256 / FIPS 180-4](#). These guide selected checks; they do not establish full standards conformance.

## 2. Test Coverage and Reproducible Code

| Test group                | Checks and purpose                                                                                                                                                                    |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Access and health         | Check login, origin restrictions, and DataLink/Streamer ping and health endpoints to establish protected access and service availability.                                             |
| DID and address discovery | Resolve four file DIDs and one dataset DID; parse VOTables and verify advertised direct-file and Streamer URLs. Bootstrap addresses are configured, not SCAPI-discovered.             |
| File and dataset delivery | Download four files through direct links and Streamer; compare byte counts and independently generated SHA-256 values. Inspect all three TAR members without extracting them to disk. |
| Partial data and errors   | Verify exact 100-byte retrieval and HTTP 416 rejection; test missing-token, denied and wrong-audience synthetic identities, malformed XML, and blocked mock-token access.             |
| Diagnostics and evidence  | Inspect MIME, standardID and authentication discovery; retain address provenance, responses, downloaded fixtures and machine-readable results.                                        |

**Public GitHub repository:** [github.com/guotsuan/datalink-streamer-lab](https://github.com/guotsuan/datalink-streamer-lab).

The repository includes server integration code, systemd examples, dependency snapshots, Python tests and an English [README.md](#). [Deployment instructions](#) distinguish the original VM configuration from fresh setup. Run the [external test script](#) with `python3 lab/test_public_e2e.py`; enter the disposable password when prompted. Passwords, signing keys and local runtime evidence are excluded from the repository.

## 3. Test Results

**Latest supplied external run: 22 September 2026, 15:05:15 UTC.** A total of **41 HTTP requests** produced **33 checks: 30 PASS / 3 WARN / 0 FAIL**. The internal suite exported by this run returned **14/15**, with one MIME diagnostic failure overlapping an external warning. These counts match the 21 September baseline; requests, checks and suites have different denominators.

**Verified outcome.** DID-to-link resolution, four-file delivery, three-member TAR integrity, exact partial bytes and the tested negative responses worked with simulated dependencies. Latest evidence run:

20260922T150508Z-661cd650. Pages 3-6 contain supplied screenshots, all 33 check results, file hashes and an address inventory. The full original `report.md` is embedded as a PDF attachment.

**Recorded observations.** Three diagnostic warnings remain: DataLink returns `application/xml`; DataLink standardID INFO is absent; and the public unauthorized response omits WWW-Authenticate and advertises a localhost mock discovery URL. These observations are retained, not relabelled as passes.

**Conclusion.** The tests establish a repeatable synthetic software-verification baseline, not production readiness or full IVOA certification. Real discovery/authentication, interrupted-transfer recovery, large-file/concurrency tests and failure recovery remain unverified. Partial-download testing uses Streamer POST, not standard HTTP GET-range conformance. This revision incorporates the user's new evidence; no additional live tests were executed during report preparation.

## 3.1 Portal and DataLink Response

OPERABILITY / MINIMAL TEST ENVIRONMENT

# Follow the link. Verify the bytes.

A small, inspectable path from a data identifier to a verified file.  
Official SKA DataLink and Product Streamer, running on your VM.

INPUT

Synthetic DID  
Local fixture catalogue

→

DISCOVER

DataLink UP  
Official VOTable response

→

DELIVER

Streamer UP  
Official byte / TAR stream

→

VERIFY

SHA-256  
Independent fixture manifest

MANUAL EXPLORER

### Resolve a product

Data identifier (DID)

Single FITS   3-file dataset   1 MiB file

Simulated identity

Allowed laboratory identity

Range header optional; single file only

Resolve DataLink Download via Streamer

Descriptor found. Send this VOTable to Streamer to test delivery.

身份选项仅使用公开的测试令牌，不输入真实 SKA 凭据。下载只用于少量测试文件。

Response inspector

DataLink HTTP 200

Discovered accessURL: <http://64.176.188.89:18080/streamer/v1/data/product>

```
<?xml version="1.0" encoding="UTF-8"?>
<VOTABLE xmlns="http://www.ivoa.net/xml/VOTable/v1.3"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" version="1.4">
  <RESOURCE type="results">
    <TABLE>
      <FIELD name="ID" datatype="char" arraysize="*" ucd="meta.id;meta.main" />
      <FIELD name="access_url" datatype="char" arraysize="*" ucd="meta.ref.url" />
      <FIELD name="service_def" datatype="char" arraysize="*" ucd="meta.ref" />
      <FIELD name="error_message" datatype="char" arraysize="*" ucd="meta.code.error" />
    />
    <FIELD name="semantics" datatype="char" arraysize="*" ucd="meta.code" />
    <FIELD name="description" datatype="char" arraysize="*" ucd="meta.note" />
    <FIELD name="content_type" datatype="char" arraysize="*" ucd="meta.code.mime" />
    <FIELD name="content_length" datatype="long" ucd="phys.size;meta.file" />
    unit="byte" />
    <FIELD name="content_qualifier" datatype="char" arraysize="*" ucd="meta.code" />
  </DATA>
</VOTABLE>
```

Figure 1. User-supplied portal capture: lab.test:image.fits resolves with HTTP 200, a VOTable 1.4 response and the public Streamer accessURL. The visible labels identify simulated dependencies. No capture timestamp is shown; this image alone does not establish successful file delivery.

## 3.2 Internal Suite Screenshot

REPEATABLE EVIDENCE

Run the minimum checks

[Download JSON report](#) [Run test suite](#)

Checks the actual service processes, not a canned result. Standards diagnostics may fail: upstream differences are preserved and reported.

14 / 15 checks passed · 2026-09-22T15:03:24.248274+00:00. Failures remain visible for investigation.

| CHECK                                    | RESULT | HTTP | EVIDENCE                                                                |
|------------------------------------------|--------|------|-------------------------------------------------------------------------|
| DataLink ping                            | PASS   | 200  | HTTP 200                                                                |
| DataLink health                          | PASS   | 200  | HTTP 200                                                                |
| Streamer ping                            | PASS   | 200  | HTTP 200                                                                |
| Streamer health                          | PASS   | 200  | HTTP 200                                                                |
| Resolve single-file DID                  | PASS   | 200  | Parseable V0Table; Streamer accessURL discovered                        |
| Resolve dataset DID                      | PASS   | 200  | Parseable V0Table; Streamer accessURL discovered                        |
| V0Table → file bytes + SHA-256           | PASS   | 200  | SHA-256 960bdf5cafd024bce6d14e10c11b9dee8d35efc0b328dfc6080080628c18600 |
| No token → 401                           | PASS   | 401  | HTTP 401                                                                |
| Denied fixture token → 403               | PASS   | 403  | HTTP 403                                                                |
| Dataset → TAR + member integrity         | PASS   | 200  | 3 TAR members; sizes and bytes match fixture manifest                   |
| JSON request + Range 0–99                | PASS   | 206  | 206; exact 100 bytes and Content-Range                                  |
| Unsatisfiable Range → 416                | PASS   | 416  | HTTP 416                                                                |
| Missing file → 404                       | PASS   | 404  | HTTP 404                                                                |
| Malformed XML → 400                      | PASS   | 400  | HTTP 400                                                                |
| DataLink standards baseline (diagnostic) | FAIL   | —    | Upstream returns application/xml, not DataLink MIME type                |

Figure 2. User-supplied internal-suite capture at 22 September 2026, 15:03:24 UTC: 14/15 checks pass, with the DataLink MIME diagnostic retained as FAIL. This is a separate run from the external report completed at 15:05:15 UTC; it is not the same execution or denominator.

### 3.3 External Test Check Register

Source: report.md / report.json, run 20260922T150508Z-661cd650. Completed 22 September 2026, 15:05:15 UTC. 41 requests; 33 checks: 30 PASS, 3 WARN, 0 FAIL.

| No. | Result | Recorded check                                                    |
|-----|--------|-------------------------------------------------------------------|
| 1   | PASS   | Anonymous homepage redirects                                      |
| 2   | PASS   | Anonymous metadata denied                                         |
| 3   | PASS   | Login with same-origin form and open homepage                     |
| 4   | PASS   | Reject null-origin login                                          |
| 5   | PASS   | Reject foreign-origin login                                       |
| 6   | PASS   | Reject cross-site login                                           |
| 7   | PASS   | Catalogue and independent fixture hashes                          |
| 8   | PASS   | Aggregated service status                                         |
| 9   | PASS   | Public endpoint /datalink/v1/ping                                 |
| 10  | PASS   | Public endpoint /datalink/v1/health                               |
| 11  | PASS   | Public endpoint /streamer/v1/ping                                 |
| 12  | PASS   | Public endpoint /streamer/v1/health                               |
| 13  | PASS   | Discover lab.test:image.fits and verify advertised direct URLs    |
| 14  | PASS   | Streamer download and integrity: image.fits                       |
| 15  | PASS   | Discover lab.test:readme.txt and verify advertised direct URLs    |
| 16  | PASS   | Streamer download and integrity: readme.txt                       |
| 17  | PASS   | Discover lab.test:metadata.json and verify advertised direct URLs |
| 18  | PASS   | Streamer download and integrity: metadata.json                    |
| 19  | PASS   | Discover lab.test:range.bin and verify advertised direct URLs     |
| 20  | PASS   | Streamer download and integrity: range.bin                        |
| 21  | PASS   | Discover lab.test:dataset and verify advertised direct URLs       |
| 22  | PASS   | Streamer download and integrity: dataset                          |
| 23  | PASS   | Range download: exact 100 bytes                                   |
| 24  | PASS   | Out-of-bounds Range rejected                                      |
| 25  | PASS   | Missing access token                                              |
| 26  | PASS   | Denied identity                                                   |
| 27  | PASS   | Wrong token audience                                              |
| 28  | WARN   | Authentication discovery usability                                |
| 29  | PASS   | Malformed XML rejected                                            |
| 30  | PASS   | Mock token endpoint not publicly exposed                          |
| 31  | WARN   | DataLink MIME diagnostic                                          |
| 32  | WARN   | DataLink standardID INFO diagnostic                               |
| 33  | PASS   | Run internal suite and verify exported report                     |

The final PASS means the internal report was retrieved and its known diagnostic failure preserved; it does not mean all 15 internal checks passed. Full per-check evidence is retained in the embedded original report.md.

### 3.4 File Integrity and Address Inventory

| Fixture / bytes       | SHA-256 (independent expected bytes and downloaded file)             |
|-----------------------|----------------------------------------------------------------------|
| image.fits / 5,760    | 960bdf5cafd024bce6d14e10c11b9de<br>e8d35efc0b328dfc6080080628c18600  |
| readme.txt / 58       | d62a016c10fbb83146ef19e26b5e7238<br>751cde4b10f7816e43a1bf5c043049c3 |
| metadata.json / 39    | 96760fdaf6d9b0eace5b26183e9af7a7<br>c43fe1bb0e706004038d55664ee2dc12 |
| range.bin / 1,048,576 | fbbab289f7f94b25736c58be46a994c4<br>41fd02552cc6022352e3d86d2fab7c83 |

The three-member TAR contains image.fits, readme.txt and metadata.json. TAR size: 9,728 bytes. The partial download contains exactly 100 bytes with Content-Range: bytes 0-99/1048576.

Public base: <http://64.176.188.89:18080>. Repeated address records are consolidated below; all original provenance entries remain in the embedded report.

| Address (relative to public base unless shown in full)                                                                                  | Provenance / use             |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| /                                                                                                                                       | Configured entry point       |
| /login                                                                                                                                  | Configured entry point       |
| /lab/info                                                                                                                               | Configured entry point       |
| /lab/status                                                                                                                             | Configured entry point       |
| /datalink/v1/links                                                                                                                      | Configured entry point       |
| /datalink/v1/ping                                                                                                                       | Configured entry point       |
| /datalink/v1/health                                                                                                                     | Configured entry point       |
| /streamer/v1/ping                                                                                                                       | Configured entry point       |
| /streamer/v1/health                                                                                                                     | Configured entry point       |
| /lab/run                                                                                                                                | Configured entry point       |
| /streamer/v1/data/product                                                                                                               | VOTable descriptor           |
| /fixtures/lab.test/image.fits                                                                                                           | VOTable row                  |
| /fixtures/lab.test/readme.txt                                                                                                           | VOTable row                  |
| /fixtures/lab.test/metadata.json                                                                                                        | VOTable row                  |
| /fixtures/lab.test/range.bin                                                                                                            | VOTable row                  |
| <a href="http://localhost:18080/mock/.well-known/openid-configuration">http://localhost:18080/mock/.well-known/openid-configuration</a> | Error response; NOT followed |
| /lab/reports/a96514d0-c5d1-4726-9b24-3f0900c169a9.json                                                                                  | /lab/run response            |

Evidence scope: configured bootstrap routes plus approved same-origin data links, not a SCAPI discovery test. Mock IAM discovery is not followed. HTTP is unencrypted; only synthetic data and disposable credentials are in scope. Full original report.md is available through the PDF attachment panel in compatible readers.